

Matgeo-4.13.88

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Question

A line with positive direction cosines passes through the point $P(2, -1, 2)$ and makes equal angles with the coordinate axes. The line meets the plane $2x + y + z = 9$ at point Q . Find the length of PQ .

Solution

Step 1: Line in vector form

$$(P) = \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix}, \quad (d) = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \quad (1)$$

$$(r) = (P) + t(d) = \begin{pmatrix} 2+t \\ -1+t \\ 2+t \end{pmatrix} \quad (2)$$

Step 2: Plane in vector form

$$(n)^T (r) = 9, \quad (n) = \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix} \quad (3)$$

Solution

Step 3: Intersection point

$$(n)^T ((P) + t(d)) = 9 \quad (4)$$

$$(2 \ 1 \ 1) \left(\begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix} + t \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \right) = 9 \quad (5)$$

$$\underbrace{(2 \ 1 \ 1) \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix}}_{=5} + t \underbrace{(2 \ 1 \ 1) \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}}_{=4} = 9 \quad (6)$$

$$5 + 4t = 9 \Rightarrow t = 1 \quad (7)$$

$$(8)$$

Solution

$$(Q) = (P) + t(d) = \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix} + 1 \cdot \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 0 \\ 3 \end{pmatrix} \quad (9)$$

$$(PQ) = (Q) - (P) = \begin{pmatrix} 3 \\ 0 \\ 3 \end{pmatrix} - \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \quad (10)$$

$$|(PQ)| = \sqrt{(PQ)^T (PQ)} = \sqrt{1^2 + 1^2 + 1^2} = \sqrt{3} \quad (11)$$

Conclusion

Conclusion:

$$\overrightarrow{PQ} = (Q) - (P) = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \quad (12)$$

$$|PQ| = \sqrt{1^2 + 1^2 + 1^2} = \sqrt{3} \quad (13)$$

Therefore, the length of the line segment PQ is $\sqrt{3}$.

Graphical Representation

